2 The demand for health and health services

2.1 Demand and demand for health care

In most parts of the economy individuals make choices about what goods and services they wish to consume, and express these demands in the market. With their knowledge about what goods they like, how much they like them, and knowledge about what is available and at what price, they make choices. The theory of demand is about trying to understand how people make choices and how their choices are likely to change with changing circumstances.

This chapter outlines the economic theory of demand, and considers its application in the fields of health and health care. Some of the issues will be considered in much more detail in later chapters. In particular we will discuss the many ways in which markets do not work well in health and health care, but at this stage the focus is on the main factors that influence choices.

2.2 What are the main factors that affect choices?

In many of our decisions on what to buy and how much to buy the key consideration is price. At lower prices we are more likely to buy a good, and are more likely to buy larger quantities. This relationship is often presented diagrammatically in a *demand curve*, which shows that the quantity demanded tends to be lower at higher prices and higher at lower prices (see Figure 2.1).

More people will choose to buy a good at lower prices and those who do buy will tend to buy more. There may be limits to how much of any particular good people will buy. For example, people will only take pain killers when they have a pain, and even then they can only take limited quantities. There is an underlying principle here – the more we have of any particular good the less we value additional quantities (sometimes called the law of diminishing marginal utility). There may come a point where we simply do not want any more (that is the extra benefit we expect from consuming more is zero or negative). However, in general we demand more at lower prices and less at higher prices. This is represented in Figure 2.1 as moving down the demand curve.

The amount of any good we demand also depends on what are the alternatives. If we want a pain killer we may have the choice between ibuprofen, paracetamol and ASA. Demand for ibuprofen will be lower when ASA is cheap and higher when ASA is expensive. Demand therefore depends on price, but also the price and availability of (perfect or near) substitute goods. The extent to which the price of substitutes affects demand depends on how close a substitute it is. In the example here we might expect that the price of ASA is more likely to

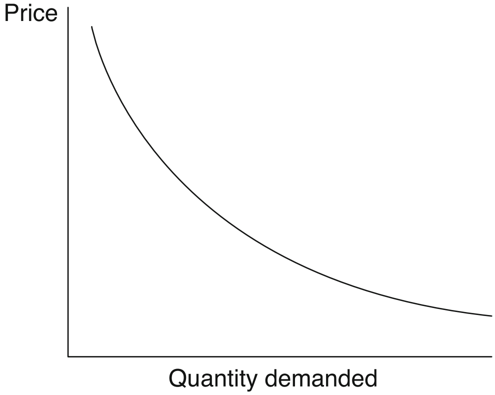

Figure 2.1 The demand curve

affect demand for ibuprofen than would be the case for paracetamol since it is a more similar type of drug. However, there are some circumstances where the choice between ibuprofen and paracetamol might depend on their relative prices.

When the price of a substitute good falls this reduces demand for the original good (at any price). This can be represented as a shift in the demand curve from D1 to D2, as shown in Figure 2.2. A common example of this is when a drug goes off patent. Other producers can now come into the market with generic versions of the drug so the price is likely to fall. Older (sometimes less effective) drugs may find that demand falls as people (or health system organisations) can now buy the better drug at a lower price.

Demand for a good is also affected by the price and availability of complementary goods – that is ones that are used in combination with the good. For example, if the price of fuel rises this may reduce the demand for cars. The effect is the opposite to what is seen with substitutes. When the price of a complementary good rises that will reduce demand for the original good (at any price). In essence people are buying the bundle of cars and fuel, and if fuel prices rise this increases the price of the bundle.

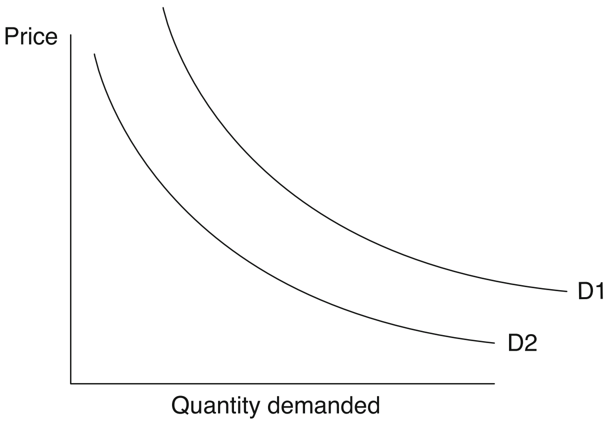

Figure 2.2 A shift in the demand curve

A fourth important factor is income. Richer people can buy more of everything, so that demand for most goods and services will rise with higher incomes and fall with lower incomes. In many instances richer people buy more of particular goods, but in some cases they may buy less as they switch to better quality alternatives. For example there may be a reduction in the use of bus travel and an increase in the use of taxis when individuals get richer. The effect of income applies both at the individual level and across individuals – an overall rise in the income of a country is usually associated with more demand for goods and services. In terms of the diagram in Figure 2.2, the effect of a rise in income is likely to shift the demand curve from D2 to D1. If demand increases as incomes increase, this is defined as a 'normal good'. If demand falls as incomes rise, this is called an 'inferior good'.

In summary, the factors that are easily observable and affect demand for a good are its own price, the price and availability of substitute goods, and the price and availability of complementary goods and incomes. Demand falls as price rises, falls when the price of substitutes falls, falls when the price of complements rises, and falls as incomes fall. Demand increases as price falls, the price of substitutes rises, the price of complements falls and when incomes rise.

2.3 Other factors that affect choices

The factors discussed above are relatively easy to observe. But there are other, less visible ways in which our preferences are formed and our choices made. Many of our tastes are formed by habit – the preference for eating rice, potatoes or maize may reflect the fact that one of these has always been plentiful locally, and people get into the habit of eating it. Local climate may influence the types of food people choose and the clothes they wear, and whether or not they buy heating fuel.

Some reasons for choices may reflect access to information or propaganda. For example, the main purpose of advertising is to change people's views and choices. This may encourage people to consume more, or to consume particular brands. Successful advertising will shift the demand up at any price – a shift from D1 to D2 in Figure 2.2. To some extent all people are affected by fashions. It can be difficult to predict the next fashions, but they can have substantial effects on patterns of demand. Evidence linking certain products to improved or worse health may affect preferences. For example, the publication of evidence that smoking reduces life expectancy might reduce demand for cigarettes at any price. This can be represented as a shift in the demand curve from D2 to D1 in Figure 2.2.

2.4 How is the demand for health care different?

There are some obvious ways in which the way we interact with the health sector is different from our dealings with other providers of goods and services. Doctors advise us on what services we need and often also provide them. This means that the individual making the choices may not be the person using the service, or at least there may be a joint decision between the doctor and the patient. The problem is that the patients may not be very well informed about the illness and the treatment choices, so they need help in making decisions. Similar problems arise in the markets for other professional services – people typically know less about the law than the lawyers who advise them. As will be discussed more fully in later chapters, demand for health and health care may reflect in part the information held by health care professionals and their views, and the ways in which professionals are better informed than users of the services.

Some health services are used when we are very ill and may not be able to make sensible decisions. Some health care decisions are literally about life and death. Again this means that to an extent choices will be made on behalf of the service user, either by professionals or by family and friends. Even when the patient is able to understand his or her needs, and how the treatment is likely to improve health, there can be problems in making choices. In many cases outcomes of treatment are very uncertain, and people find it difficult to make choices when faced with very uncertain results.

Demand for any good reflects both the preferences of individuals, and their ability to find resources to pay for it. However serious the medical problem, if someone cannot afford the treatment, and cannot persuade anyone else to pay, then the need is not translated into a visible demand. In practice in the area of health care there are often other parties who will pay, either through insurance or some form of government or charitable funding. This means that the direct user of services might not really care about the price, and indeed may be pleased to get an excellent service at a high price. When we pay insurance, so long as the patient does not have to pay for treatment, there may be little reason to respond to price signals. There are many good reasons why health services are normally provided (at least in part) through insurance or other third party payers, but an inevitable consequence is to reduce the effect of price on demand for services.

Another problem is timing. In general we are healthier when relatively young and relatively rich. These are times when we are least likely to need health care, but are most able to afford it. In general we have medical needs when we are older and usually poorer. This pattern of needs and incomes means that mechanisms are needed to allow us to pay when we can, and use services when we need them. Most such mechanisms break the direct link between demand and price, since the user faces only part or none of the price charged.

Perhaps the most important feature of our need for health care is that we seldom know in advance what we will need, when we will need it or how much we will need. This uncertainty means that some form of insurance is needed (since the costs of some treatments would not be affordable unless insured). Again the effect of insurance can be to break the link between the decision to use a service and the price charged.

In contrast to many goods and services, we seldom actually enjoy using health care. What we are typically buying is long term gains at the expense of short term pain or inconvenience. This contrasts to many other goods and services – most people enjoy eating as well as enjoying the reduction in hunger. Demand for health care can therefore reflect some reluctance to put up with the short term pain for (possibly uncertain) long term gains.

2.5 In what ways is demand for health care like demand for other goods?

On the other hand not all health interventions are uncertain, few are really about life and death, and in many cases the intervention is well understood by the patient. For example, if you have myopia you need optometry services. You can calculate with almost perfect accuracy how often you need eye tests and, unless you sit on them, how many pairs of spectacles you will need for the rest of your life. For many people dental care is almost as predictable. There is no significant uncertainty in the need for many childhood vaccinations – the content and timing of immunisation are predictable.

Many health services are about comfort, mobility, feeling healthy and having good quality of life. Relatively little of what is done extends life to a significant degree. In an absolute sense health care is less necessary than many other necessities, such as food and clothing.

While markets may not be the best way to allocate some health care resources, for many health services people's behaviour reflects normal patterns of demand. People buy more at lower prices and when income rises. They buy less at higher prices or when income falls. Even where health care markets are replaced with other resource allocation systems the patterns of demand may be important. If the price facing the user is low or zero then demand will tend to be high, and this can explain long queues and waiting times for treatment.

There are several reasons why we should be interested in demand for health and health care. The first is to help us to predict likely reactions and behaviour. For example, if we charge people a fee for eyesight tests, what will be the effect on the number of people using the service? How will such a charge affect the frequency of use of optometry services? Second, knowing something about people's demand for health care may tell us something about how much they value services. This point will be explored in greater depth below.

2.6 Demand functions

In this model of demand, the quantity chosen is a function of the price:

$$Q = f(P)$$

Those familiar with mathematical convention will notice that demand curves have been drawn with price on the vertical axis and quantity on the horizontal axis. This is the opposite of the normal convention in mathematics for the dependent and independent variable. Alfred Marshall, who first suggested this analysis in the nineteenth century, used the axes in this way, and the habit has stuck.

A fuller description of the determinants of demand includes the other factors discussed. More formally we can express this as

$$D = f(P, Ps, Pc, Y, T)$$

where P is price, Ps is the price of substitute goods, Pc is the price of complement goods, Y is income and T is tastes.

While it is easy to specify what factors affect demand, it is an empirical question in what ways these factors affect demand. The demand curves drawn in Figures 2.1 and 2.2 could take many shapes, but in general will slope downwards. The exact relationships can be estimated statistically, and evidence from demand studies can be useful in understanding how demand will respond to changing circumstances.

2.7 Demand, elasticity and health

This chapter has been concerned with understanding the factors that affect demand for goods generally and for health in particular. A key issue is how much changes in these factors affect demand. Economists measure the effects using the concept of *elasticity*. Elasticity measures the responsiveness of demand to changes in prices, incomes or the prices of substitute or complement goods. The most commonly estimated elasticities are for price and income.

It is often useful to know more about how demand varies with price. Some services or goods are so necessary to life that we are likely to use them whatever the price. Others bring benefits, but are less necessary or have more substitutes. The measure of how responsive demand is to price is known as the *price elasticity of demand*. In a similar way we can calculate the responsiveness of demand to income changes, and this is the *income elasticity of demand*.

2.7.1 Measuring elasticity

Many studies have estimated demand elasticities. It is often useful to know what will happen if prices rise or fall. Take the example of legalising the use of heroin. The current price of heroin is high, largely as a result of the prohibition of production, importation and distribution of the drug. If the drug were legalised it is likely that the price would fall, since it is very cheap to produce. It is very likely that consumption would rise in that case, since demand normally increases as price falls. The question of interest is ‘How much is demand likely to rise in response to a fall in price?’ Policy makers would be keen to know the answer to this before embarking on legalisation. It might be argued that, since heroin is addictive, users are not very sensitive to price. In other words demand is *inelastic*. But it would be useful to have estimates of elasticity before making the policy change.

Figure 2.3 illustrates the relationship between price and quantity of heroin demanded, per week in one market. From the information in Figure 2.3 we can draw up a table showing the relationship between quantity consumed and price. For example, at price €80, the demand would be 59 g. Table 2.1 shows all the combinations.

From the table, what can be said about the response of demand to price? The fall in quantity seems quite large – the price rise from €20 to €80 reduces demand by 31 g. But really to answer the question we need to compare the *proportionate* changes. If price falls from €80 to €60, this 25 per cent fall leads to a rise in consumption of less than 7 per cent. This suggests that demand is not very responsive to price – what might be expected from the addictive nature of the drug. If we consider a fall in price from €40 to €20, this 50 per cent fall in price leads to a 29 per cent increase in demand. In this case we can see that demand is more sensitive to price changes at low levels of price, but is very insensitive at higher prices. This is important – we often find that responsiveness changes at different levels of price.

Clearly this comparison of percentage increases in demand and decreases in price provides a measure of how much demand responds *proportionately* to price changes. This is the basis of the calculation of the price elasticity of demand. The normal way to calculate this is to

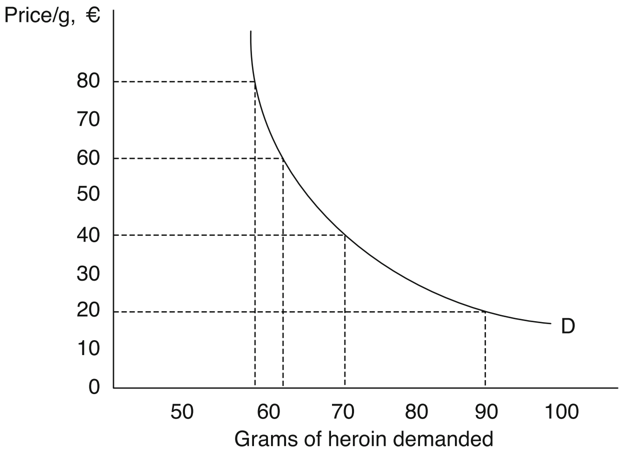

Figure 2.3 Demand for heroin

Table 2.1 Relationship between quantity consumed and price

Price (€ per gram)	20	40	60	80
Quantity of heroin sold (grams)	90	70	63	59

divide the *proportionate change in the quantity* by the *proportionate change in price*. More formally,

$$\text{Price elasticity of demand} = \frac{\text{Proportionate change in quantity}}{\text{Proportionate change in price}}$$

$$\varepsilon = \frac{\%\Delta Q}{\%\Delta P}$$

where ΔQ is the change in quantity and ΔP the change in price. In the above example, when the price changes from €80 to €60

$$\varepsilon = \frac{\%\Delta Q}{\%\Delta P} = \frac{-7\%}{25\%} = -0.28$$

By the same calculation, when the price falls from €40 to €20, the elasticity of demand is calculated at −0.58. It is clear from the calculation that elasticity is just a number – it has no units. Note also that in these cases the price elasticity of demand is a negative number. In the above example percentages were used, but of course exactly the same answer comes if proportions are used instead.

The price changes in the above example are large (we are calculating elasticity over an *arc* in the demand curve). The calculations have been carried out on the basis of lowering the price. If instead the price were raised from €20 to €40, then the percentage rise is 100 per cent. In this case it therefore matters if we are calculating the elasticity on the basis of a rise or a fall in the price. There are two ways in which we can get round this. First, the percentage rise or fall can be calculated on the basis of the mid-point of the change. If we did this we would get a slightly different estimate for elasticity:

$$\varepsilon = \frac{\%\Delta Q}{\%\Delta P} = \frac{-4}{61} \Big/ \frac{20}{70} = -0.23$$

If we are dealing with large changes in price or quantity it is preferable to calculate elasticity in this way. However, the better solution is to consider smaller changes. We could consider changes in price of €1, and there would be much less difference between the calculation using the mid-point and the calculation using the starting price to calculate the percentage.

We can easily rewrite the expression for elasticity in two parts – the ratio of the changes, and the ratio of the price and quantity:

$$\varepsilon = \frac{\%\Delta Q}{\%\Delta P} = \frac{\Delta Q}{\Delta P} \times \frac{(P2 + P1)/2}{(Q2 + Q1)/2} = \frac{\Delta Q}{\Delta P} \times \frac{(P2 + P1)}{(Q2 + Q1)}$$

where P1 and P2 are the initial and final prices, and Q1 and Q2 the initial and final quantities.

The first expression approximates the slope of the demand curve. As we make the change in price smaller and smaller we can rewrite the expression for elasticity in terms of the slope of the demand curve and the ratio of price and quantity. Using normal calculus notation:

$$\varepsilon = \frac{dQ}{dP} \times \frac{P1}{Q1}$$

This expression is for the elasticity of demand for the good or service at the point P1, Q1 (and this is known as *point elasticity*). It is now clear that elasticity can vary along the demand curve, either because of changes in slope or as a result of the changing ratio of price and quantity. Even if the demand curve is a straight line, elasticity varies along the curve. This is important to note – a flatter demand curve is normally more elastic, and a steeper one less elastic, but elasticity is not the same as slope.

In the example above the elasticity of demand was calculated to be in the range from −0.28 to −0.58. In describing price elasticity of demand we normally describe the range 0 to −1.0 as inelastic – that is to say, demand responds to changing price by a smaller proportion than the change in price. When elasticity is −1 the proportionate changes in price and quantity are the same, a situation of 'unitary elasticity'. The range from −1 to $-\infty$ is normally described as elastic. In this range the quantity demanded is highly responsive to changes in price.

2.7.2 Elasticity of demand and health promotion

Knowledge of price elasticity of demand can be useful for a range of policy decisions in health care and health promotion. An obvious example is the use of taxes to raise the price of cigarettes and discourage smoking. In this case several questions are relevant. First, if prices rise, to what extent will that achieve a reduction in smoking? Second, who will be dissuaded – will it be the young or the old, heavy smokers or occasional ones? What will happen to government revenue? Will the tax fall disproportionately on the poor?

Many studies have been carried out on the demand for cigarettes (Townsend 1987; Trigg and Bosanquet 1992; Stebbins 1991; DeCicca *et al.* 2008). Estimates vary, but the consensus is that the overall price elasticity of demand is around −0.5, but this varies greatly across different parts of the population. Using this information we can make a number of judgements about the likely effects of tax changes on smoking and tax revenue, and examine how the burden will fall on different population groups.

Let us assume that the current tax on cigarettes is an *ad valorum* tax (i.e. a percentage added to the selling price), and is set at 20 per cent of the selling price. What will be the effects of increasing it to 30 per cent of the selling price? Not all of this increase will be passed on to the consumer – the relative burden on consumer and seller depends on a range of factors. However, for simplicity let us assume that this tax is all passed on to the consumer. Since demand is inelastic, in order to produce any given percentage fall in consumption there must be a more than proportionate increase in price. Put another way, in order to produce a large reduction in smoking it is necessary to increase tax (and therefore price) substantially.

What happens to the tax revenue for the government? A simple example is set out in Table 2.2. The price increase is 8 per cent, and the fall in consumption 4 per cent. It can be seen from the table that spending has gone up despite the fall in consumption, and that tax

Table 2.2 Effect of raising the tax on cigarettes

Pre tax price (€)	*Tax (€)*	*Consumption*	*Total spending (€)*	*Tax revenue (€)*
10	2	100,000	1,200,000	200,000
10	3	96,080	1,249,040	288,240

revenue takes a larger share of the larger total spending. A simple rule of thumb is that, if demand is inelastic, a price increase caused solely by a higher tax *will always increase tax revenue*. It is sometimes suggested that governments are afraid to increase tax on tobacco for fear of losing revenue. They are almost certainly wrong. The example may understate the scope for government to increase revenue by increasing tobacco tax. In this example we have ignored the fact that normally some part of the tax is paid by the seller, so that government can increase its tax revenue from both buyers and sellers.

Studies of the elasticity of demand for tobacco have normally shown that demand is more elastic among young smokers, many of whom are recent starters. This means that the overall elasticity estimates are likely to imply that demand elasticity is even lower for older, more addicted, smokers. This would suggest that tax increases may be a good strategy for reducing smoking in younger people. However, although richer people may display lower price elasticity of demand than poorer ones, the effect of the tax is likely to fall more heavily on poorer people. Poorer people spend a higher proportion of their income on tobacco, so the tax is *regressive*, that is, it takes a higher proportion of income from those who are poorer. Taxes are described as *progressive* if richer people pay a higher proportion of their income on the tax than do poorer people.

Another point to keep in mind is that price elasticity of demand normally varies at different prices. If we know that elasticity is −0.5 at the current price we can safely predict the effects of a relatively small tax increase. What is not so sensible is to make predictions of the effects of, say, a tax rise from 20 per cent to 80 per cent, which involves a move along the demand curve, possibly to a point where elasticity is much higher.

What can be said from this simple, although realistic, example? First, increasing tax on tobacco is likely to be an effective way of reducing smoking, but large tax increases will be needed to effect a moderate reduction. Second, the government is likely to enjoy an increase in the tax revenue as a result of the increase. Third, it is likely that this policy will be particularly effective in reducing smoking in new, relatively young smokers. Fourth, such a policy may have undesirable effects on the tax burden, since it falls disproportionately on poorer people. In some countries this is an even greater issue, since smoking rates are often higher in lower-income groups.

Low price elasticity of demand tends to indicate that the good is considered to be essential, and that there are few if any substitutes. Goods may be considered essential either because they are necessary to sustain life, or as a result of addiction. One indicator that a good is addictive is that it is not normally considered necessary to sustain life but nevertheless has a low price elasticity of demand.

High taxes on health-damaging goods may be effective in reducing consumption. We can do similar calculations to investigate the effectiveness of subsidies to encourage healthy behaviours. For example, it would be possible to exempt all fresh fruit and vegetables from tax to encourage more consumption. Sports facilities can be subsidised in order to encourage use. We cannot say whether these kinds of policy will be effective ways of encouraging healthier lifestyles without doing calculations of demand elasticity.

2.7.3 Cross-elasticity of demand

Price elasticity of demand is the most widely used measure. We can also measure other elasticities. For example, we can assess the *cross-elasticity of demand* between two goods – for example, how sensitive is the demand for needles to the price of syringes? Since they are complements, we would expect demand for needles to fall if the price of needles rises. If Pn is the price of needles, and Qs is the quantity of syringes, the cross-elasticity of demand can be expressed as

$$\varepsilon = \frac{dQs}{dPn} \times \frac{Pn_1}{Qs_1}$$

If goods are complements, as in this example, the value of the cross-elasticity of demand is negative. In the case of substitutes, the cross-elasticity of demand is positive. For example, if the price of aspirin increases, we would expect a rise in demand for ibuprofen, since each can substitute for the other.

2.7.4 Income elasticity of demand

People on higher incomes can buy more of everything. Of course they do not do so – as income rises a person buys more of some goods, and less of some others. She may buy fewer bus rides and more taxi journeys, more housing and more expensive holidays. A good is said to be income-elastic if the *proportion* of income spent on the good rises with income. Income elasticity of demand can be positive (i.e. with rising income more is bought) or negative. Goods with positive income elasticity are described as *normal goods* and those with negative income elasticity as *inferior goods*.

Income elasticity is calculated from the relative change in demand for a good and the relative change in income. Formally, the formula for income elasticity of demand is

$$\varepsilon = \frac{dQ}{dY} \times \frac{Y1}{Q1}$$

where Y is the original income, and Q1 the original quantity.

2.7.5 Elasticity and prices of health care

In the same way that we can use information on elasticity to inform policy on tax or subsidy to encourage healthier lifestyles, it can also be useful in developing policy on charging for health services. In most countries there are at least some charges for medical services, hospital care or drugs. If demand is inelastic, the effect of such charges will be to raise revenue with little effect on use. If it is more elastic, charges may deter people from using effective and useful care.

The evidence suggests that in general demand for health care is price-inelastic, so that there is only a small effect on demand from raising charges (Creese 1991; McPake 1993; Gertler *et al.* 1987). However, the evidence also shows that the picture is more complicated. Poorer people display more elastic demand than do richer ones, so a simple policy of charges will deter them more. This may justify exemption from fees for people on lower incomes.

A potential role for user fees is to deter people from making unnecessary use of services. If people face a charge for access they will think carefully about their need for care. However, again the evidence suggests that such a policy cannot be applied simply. The depressing fact is that the deterrent effect of charges seems to be the same for clearly useful and for probably pointless interventions (Newhouse 1993).

Studies of the income elasticity of demand for health services suggest that the values are positive and greater than 1. In other words, demand for health care is income-elastic. As incomes rise the quantity of health care consumed rises *more than proportionately*. Goods with this characteristic are often described as luxuries, since richer people buy more. This does not imply that such goods are unnecessary, only that they are chosen in much larger amounts as income rises. This phenomenon may also explain the finding that countries that are richer not only spend more in total on health care but also spend a *higher proportion* of GNP on health services.

Overall the evidence from empirical studies of the impact of user fees suggests that they can be used to raise revenue without major deterrent effects on the use of services. However, it also shows that care has to be taken, since those on low incomes are most likely to be deterred, and are often those with the greatest needs. Understanding the patterns of elasticity of demand at different price levels and for different groups in the population allows policy to be developed to avoid some undesired effects.

Box 2.1 Estimates of elasticities of demand for health services in rural Tanzania

Sahn *et al.* (2003)[1] estimate the determinants of the demand for health services in rural Tanzania. Tanzania, like many low-income countries, has a highly pluralistic health care system (see Chapter 20) where people choose from among different types of provider – in this analysis, public and private clinics, and public and private hospitals – or they choose to seek no care at all. The analysis of Sahn and his colleagues emphasises the importance of understanding this range of options so that own price elasticities (how the demand for a particular provider changes in response to its price) and overall price elasticities (how the demand for health care in total changes in response to a change of price of one provider) can be distinguished. The net effect of price changes on the decision to seek no care at all is the suggested outcome of greatest policy relevance.

This analysis also sought to consider the role of quality of care, as well as price and income. In analysis of health care markets, quality is often difficult to observe or measure, but at the same time an important influence on decision making.

The table shows the probability of rural Tanzanians choosing particular care options, and the own and cross-price elasticities of demand between each of the options.

		Own and cross-price elasticities			
	Probability of choice	*Public hospital*	*Private hospital*	*Public clinic*	*Private clinic*
No care	0.418	0.0757	0.0563	0.0536	0.0481
Public hospital	0.057	−1.8590	0.3345	0.0795	0.0713

(Continued)

(Continued)

	Probability of choice	*Own and cross-price elasticities* Public hospital	Private hospital	Public clinic	Private clinic
Private hospital	0.05	0.4205	−1.6390	0.0795	0.0713
Public clinic	0.333	0.1116	0.0837	−0.3429	0.5826
Private clinic	0.142	0.1116	0.0837	0.6388	−1.6944
All	1	−0.0530	−0.0420	−0.0390	−0.0350

As expected, own price elasticities (on the diagonal) are negative, and cross-price elasticities are positive, indicating that the care options are substitutes. Own price elasticities are elastic for the three more expensive options (hospital care of both types and private clinic). This is probably because for each, there is a cheaper substitute available, whereas the user of a public clinic cannot respond by switching to a cheaper option when the public clinic price increases. Overall, the data imply a high degree of substitutability between options, so that when the price of one option increases, the main effect is on where people seek care, not on whether they do so. The cross-elasticity for the decision to seek no care of a change in the price of any provider is never more than 0.1, implying that were the price of that provider to double, the number of people choosing to seek no care, never increases by more than 10 per cent.

Nevertheless, it is important to distinguish between *inelastic* and *unimportant in policy terms* – a 10 per cent reduction in care seeking may be catastrophic for some. Sahn and colleagues confirm that price elasticities are larger in lower income quartiles, reaching levels of −3 for the more expensive options in the lowest income quintile (the 25 per cent of households with the lowest incomes).

	Mean own price elasticities by income quartile Quartile 1	Quartile 2	Quartile 3	Quartile 4 (highest)
Public hospital	−3.4576	−1.2552	−0.5320	−0.1344
Private hospital	−3.0454	−1.1060	−0.4745	−0.1205
Public clinic	−0.6186	−0.2445	−0.1114	−0.0310
Private clinic	−3.1610	−1.1412	−0.4740	−0.1165

They also confirm the importance of quality which had been measured only at the nearest public clinic. Interestingly, while the response of demand to quality in terms of drugs (availability) and environment (the availability of a toilet, water, and a covered waiting area) was positive only for the public clinic itself, the response of demand to quality of health staff was positive for all providers. This is probably because health staff work in more than one of the available provider types.

Note

1 Sahn, D. E., Younger, S. D. and Genicot, G. (2003) 'The demand for health care services in rural Tanzania', *Oxford Bulletin of Economics and Statistics* 65 (2): 241–59.